

NidaanKosha 100k Cohort

An exploratory analysis of cardiovascular and
diabetic risk markers across 100,000 patients in India

August 2026

DATASET

NidaanKosha-100k-V1.0

SOURCE

Hugging Face · ekacare/NidaanKosha-100k-V1.0

UNIQUE PATIENTS

100,000

PROVIDER

Eka Care

TEST RECORDS

6,844,304

PREPARED BY

Sangeeta Saha

HEADLINE FINDINGS

29.2%

high cardiovascular risk

47.4%

prediabetic or diabetic

56.0%

dysglycaemic at 41–60 yrs

1.54×

male : female lipid risk

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FIGURES Figures 1–3 describe the cohort. Figures 4–19 give the lipid distributions across the four reference-range groups. Figures 20–21 cover the cardiovascular risk ratios and Figures 22–24 fasting glucose.

Executive summary

The NidaanKosha-100k-V1.0 dataset holds 6,844,304 diagnostic test results for 100,000 patients across India, released by Eka Care. This report examines the subset of tests that speak to cardiac and metabolic health: the four components of the lipid panel, the four cholesterol ratios derived from them, and fasting plasma glucose. The cohort is not a population sample; it is everyone who walked into a laboratory and had these tests run. Read the findings below as a description of who gets tested and what those tests show, not as national prevalence.

100,000patients in cohort
6,844,304 test records**87,761**with a lipid ratio
87.8% of cohort**65,955**with fasting glucose
66.0% of cohort**25,583**high cardiovascular risk
29.2% of lipid cohort**31,243**prediabetic or diabetic
47.4% of glucose cohort**41,931**patients aged 41–60
41.9% of cohort

Seven findings

- Risk rates and risk counts point at different age groups.** Counted heads, high cardiovascular risk peaks at 41–60 years (11,611 patients). Measured as a rate, male risk peaks a full band earlier, at 21–40 years, where 35.7% of men carry a high-risk lipid ratio against 32.0% at 41–60. The 41–60 band leads on counts only because it is the largest band in the cohort. Screening designed from counts will arrive roughly fifteen years late.
- The sex gap in lipid risk is a young-adult phenomenon that closes with age.** The male-to-female high-risk rate ratio falls from 1.97 at 21–40 to 1.43 at 41–60 and 1.13 above 60. Women's lipid risk rises steadily with age while men's plateaus early, so the two converge in later life.
- By middle age, abnormal fasting glucose is the majority state.** Among tested patients aged 41–60, 56.0% are prediabetic or diabetic against 43.5% normal. The crossover from mostly-normal to mostly-abnormal happens inside this band: at 21–40 the dysglycaemic share is 27.8%, above 60 it is 63.1%.
- The prediabetes reservoir is largest exactly where it is most reversible.** At 21–40 there are 1.60 prediabetic patients for every diabetic one; by 41–60 the ratio has fallen to 0.98 and above 60 to 0.98. The pipeline empties into diagnosed disease between those bands. The 16,228 prediabetic patients are the single largest preventable-conversion pool in the dataset.
- The sex gap is far wider for lipids than for glucose.** Men carry 1.54 times the female rate of high-risk lipid ratios but only 1.22 times the female diabetes rate and 1.12 times the dysglycaemia rate. A single statement that men are at higher risk flattens a real difference between the two axes.
- Only 31.3% of the lipid-tested cohort sits at ideal ratios.** Roughly two in three patients are at least one band away from ideal. Most of that mass is in the acceptable band (39.5%, 34,688 patients), which is where lifestyle intervention still works and where no clinical flag is currently raised.

7. **The two headline percentages rest on different denominators and must not be compared directly.** Lipid ratios exist for 87.8% of patients, fasting glucose for 66.0%. The 29.2% and 47.4% figures describe different populations of different size; neither can be added to nor subtracted from the other.

READING THE AGE AND SEX SIGNAL

Read off raw counts, these data point to the 41–60 band and to male sex as the screening priorities. Both hold, but the rate-based view sharpens them: the male lipid-risk rate is already at its maximum by 21–40, and the glucose gradient is steeply age-driven while the lipid gradient is not. A screening programme built on these data would start lipid screening for men in their late twenties and run glucose screening hardest from 40 onward.

Scope

Three analytes and the four ratios derived from them, drawn out of a file holding 6.8 million results across every test in the release.

- **Examined here.** Total cholesterol, LDL, HDL and triglycerides; the four cholesterol ratios built from them; and fasting plasma glucose. Each is analysed across the whole cohort and, where reference ranges differ, separately for under-19s, adult men and adult women.
- **Present in the data, not examined.** Every other analyte in the release — liver enzymes, thyroid panels, iron studies, vitamin D and the rest. The same cohort supports all of them and the method used here would carry across unchanged.
- **Deliberately not attempted.** Any claim about national or regional prevalence, any causal statement, and any interpretation at the level of an individual patient. The reasons are set out under Limitations.

STATUS

Exploratory analysis for research and educational purposes. Not a clinical guideline, not a diagnostic instrument, and not a basis for individual patient management.

Method and data quality

Every figure in this report depends on how the cohorts were built and how inconsistent values were handled. Those decisions materially affect each number that follows, so they are set out in full before any result is presented.

Cohort construction and denominators

Each row in the source data is one test result; a patient contributes as many rows as tests they had. All analysis therefore counts distinct **document_id** values, not rows. Because patients receive different test panels, each marker has its own denominator.

Analysis cohort	Patients	Share of 100,000	Basis
Demographics (age, sex)	100,000	100.0%	every patient in the release
Lipid ratios (any of four)	87,761	87.8%	at least one ratio result, value ≤ 10
Fasting plasma glucose	65,955	66.0%	LOINC 1558-6 or 14771-0, value ≤ 400 mg/dL

Table 1 | Cohort sizes. Percentages elsewhere in this report are taken against the relevant row here, never against 100,000 unless stated.

WHY THIS MATTERS

A reader who sees “29.2% high cardiovascular risk” and “47.4% prediabetic or diabetic” will naturally place them side by side. They cannot be. The first is 25,583 of 87,761 patients; the second is 31,243 of 65,955. Roughly a third of the cohort is missing from the glucose figure and an eighth from the lipid figure, and the missing patients are not missing at random — a lipid panel and a fasting glucose test are ordered for different clinical reasons.

Unit harmonisation and outlier handling

Free-text unit fields in the source data are inconsistent to a degree that makes naive aggregation meaningless. HDL cholesterol alone arrives under 50 distinct unit strings, including **mg/dl**, **mg / dL**, **MG/DL**, **mg.dl.**, **mgs / dl**, **mg / / dL**, **mmol/L** and **umol/L**. Every unit string was lower-cased, stripped of internal whitespace and punctuation, mapped to a canonical form, and then converted to mg/dL with analyte-specific factors (38.67 for cholesterol in mmol/L, 18 for glucose in mmol/L).

Data-quality issue found	Scale	Treatment applied
Divergent unit spellings for one analyte	up to 50 variants	normalised, then converted to mg/dL
Values implying a mis-entered unit	6 total-cholesterol records above 10,000 mg/dL (max 230,000)	removed by the 400 mg/dL ceiling
Free-text reference ranges	5+ formats, e.g. 125.0 - 200.0, < 200.0, desirable <200 borderline 200-239 high >240	parsed by regular expression into numeric bounds

Data-quality issue found	Scale	Treatment applied
Several LOINC codes per analyte	17 distinct codes match "cholesterol"	codes grouped by analyte before aggregation
Missing specimen type	present across the file	imputed as the most frequent specimen for that LOINC
Implausible lipid ratios	values above 10	excluded from risk classification

Table 2 | Data-quality findings and the rule applied to each. The 400 mg/dL analyte ceiling and the ratio ceiling of 10 are blunt instruments: they remove genuine extreme values along with clerical errors.

How lipid risk was assigned

A patient may have up to four ratio results, and those results need not agree. Each ratio was classified independently against its own thresholds, after which the patient was assigned the risk band held by the largest number of their ratios. Ties were broken toward the more severe band, in the order high risk, then acceptable, then ideal. This is a deliberately conservative rule and it biases the overall picture toward higher risk: a patient with one high-risk and one ideal ratio is recorded as high risk. Readers comparing the 29.2% figure with published prevalence estimates should keep that tie-break in mind.

1 Cohort profile

Patients range from 17 to 105 years of age, concentrated between 30 and 65. Of the 100,000 patients, 57,346 are male (57.3%) and 42,654 female (42.7%). That imbalance is a property of who presents for diagnostic testing rather than of the population, and it is the reason the rest of this report reports rates within sex rather than counts across sexes.

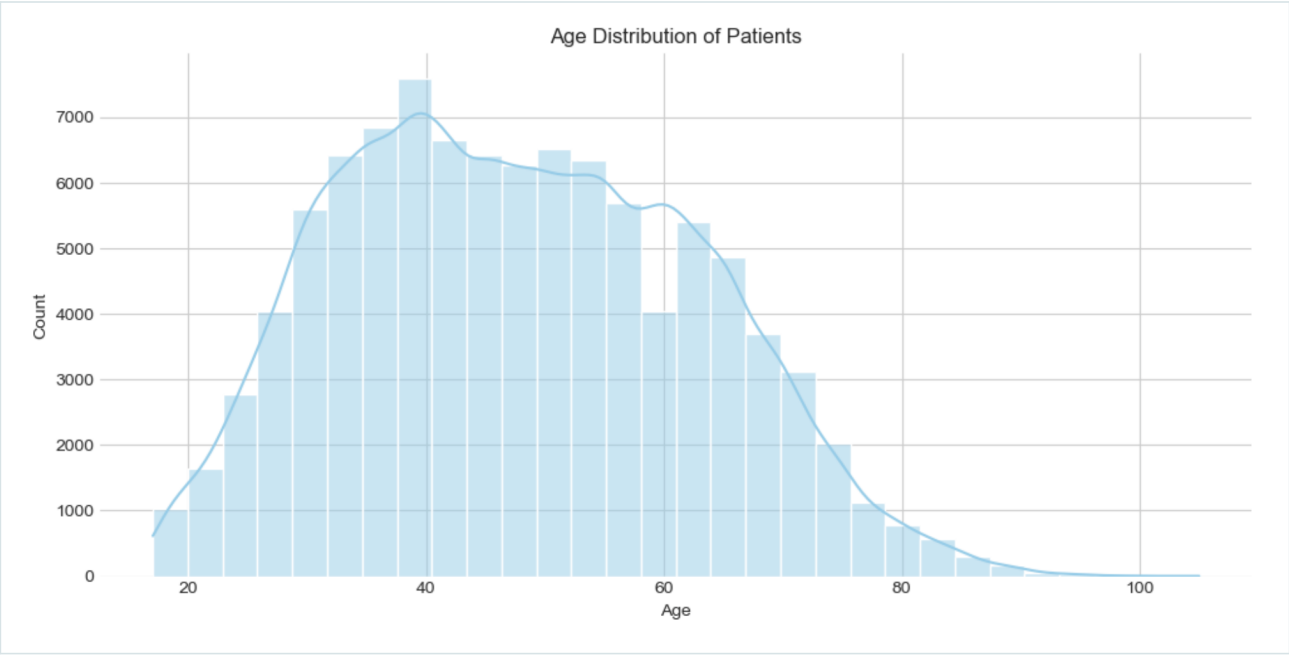


Figure 1 | Age distribution of patients, with kernel density overlay.
The distribution is broad and left-skewed toward middle age, with a visible shoulder in the thirties and a long thin tail beyond 85. Very few patients under 20 appear at all, which limits what the paediatric reference-range analysis on the following pages can support.

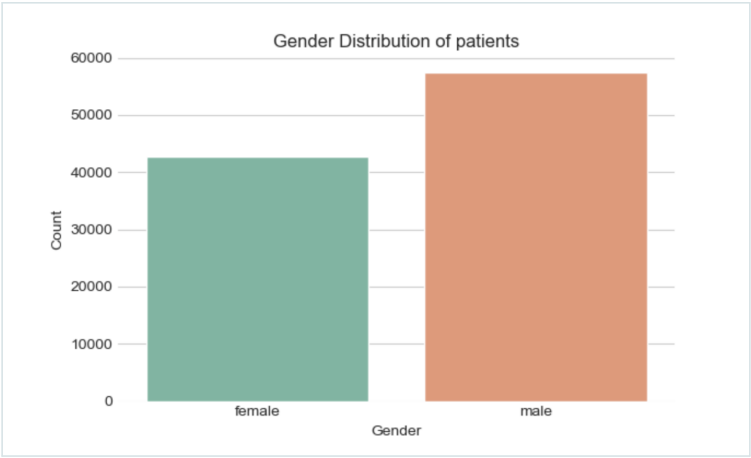


Figure 2 | Patient count by sex.
The 57:43 male-to-female split holds broadly across age bands, widening to 60:40 at 21–40 and narrowing to 53:47 at 13–20.

Age × sex structure

Age was grouped into four bands used throughout the report: 13–20 years (teens and young adults), 21–40 (adults), 41–60 (middle-aged) and above 60 (seniors). The bands are unequal in width and very unequal in population, which is why every later comparison is expressed as a rate.

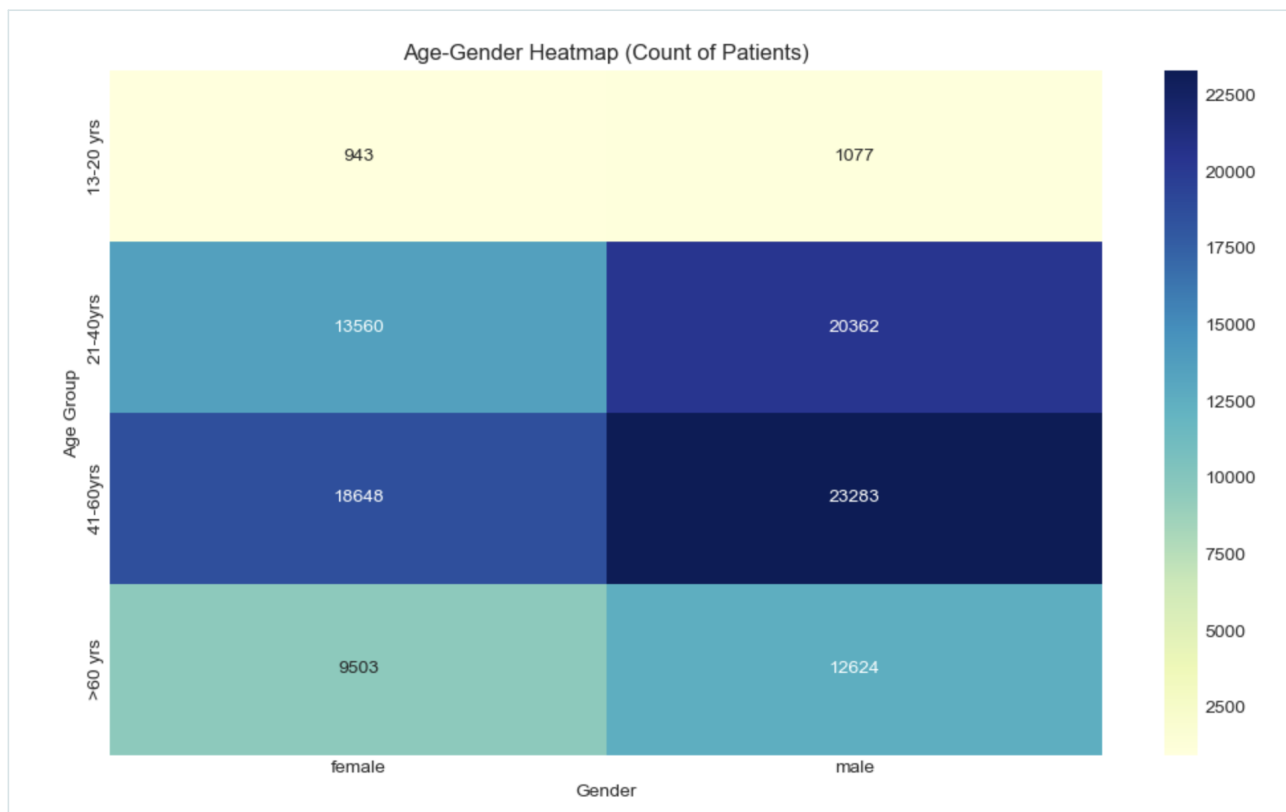


Figure 3 | Patient counts by age band and sex.

The 41–60 band is the largest at 41,931 patients, followed by 21–40 at 33,922. Together they hold 75.9% of the cohort.

Age band	Female	Male	Total	Share of cohort	Male : female
13–20 yrs	943	1,077	2,020	2.0%	1.14
21–40 yrs	13,560	20,362	33,922	33.9%	1.50
41–60 yrs	18,648	23,283	41,931	41.9%	1.25
Above 60 yrs	9,503	12,624	22,127	22.1%	1.33
All ages	42,654	57,346	100,000	100.0%	1.34

Table 3 | Cohort structure. These are the denominators used for the lipid-risk rates in Section 3.

READING THE SEX IMBALANCE

Men outnumber women in every band, but most sharply at 21–40 (1.50 to 1). Any count-based comparison between men and women therefore overstates male burden by roughly a third before any biology is involved. The corollary matters for the glucose section: fasting-glucose testing coverage is almost identical by sex (66.5% of men, 65.2% of women), so sex comparisons on that marker are unusually clean.

2 Lipid panel

Cholesterol is a fat-like substance present in every cell, necessary for building membranes and synthesising hormones. It becomes a problem in transport rather than in substance: the lipoproteins that carry it through the bloodstream differ in what they do on arrival.

What the panel measures

Component	Role	Direction
LDL — low-density lipoprotein	Delivers cholesterol to tissue; the fraction that deposits in arterial walls and builds plaque. The “bad” cholesterol.	lower is better
HDL — high-density lipoprotein	Carries excess cholesterol back to the liver for clearance. The “good” cholesterol.	higher is better
Triglycerides	Storage fat formed from unused calories; elevated levels accompany insulin resistance and independently raise cardiovascular risk.	lower is better
Total cholesterol	HDL + LDL + 20% of triglycerides. A single number that hides its own composition, which is why the ratios in Section 3 carry more information.	interpret in context

Table 4 | Lipid panel components.

Reference ranges applied

Thresholds differ by age and, for HDL, by sex. The four cohorts below correspond to the four sets of distributions in this section. All values are mg/dL.

Analyte	Under 19 yrs	Male 19+	Female 19+	All patients
Total cholesterol	< 170	< 200	< 200	< 200
LDL cholesterol	< 110	< 100	< 100	< 100
HDL cholesterol	> 45	> 40	> 50	> 40
Triglycerides	< 90	< 150	< 150	< 149

Table 5 | Reference thresholds by cohort. The dashed line on each histogram in this section marks the threshold from this table. Note the single asymmetry: the HDL floor for adult women is 50 mg/dL against 40 for men, so an identical HDL value can be normal in a man and low in a woman.

A NOTE ON THE HISTOGRAMS THAT FOLLOW

Sixteen distributions are reproduced across the next four pages, one page per cohort. Values above 400 mg/dL were removed before plotting, so each histogram is a trimmed distribution and the right tail is truncated by construction. Read the position of the bulk relative to the dashed threshold, not the extremes.

2.1 All patients

Adult thresholds applied to the full tested cohort. These four charts carry the central observation of the section: the distributions do not separate into a healthy population and a diseased one. They are unimodal, and the threshold falls close to the middle of the mass.

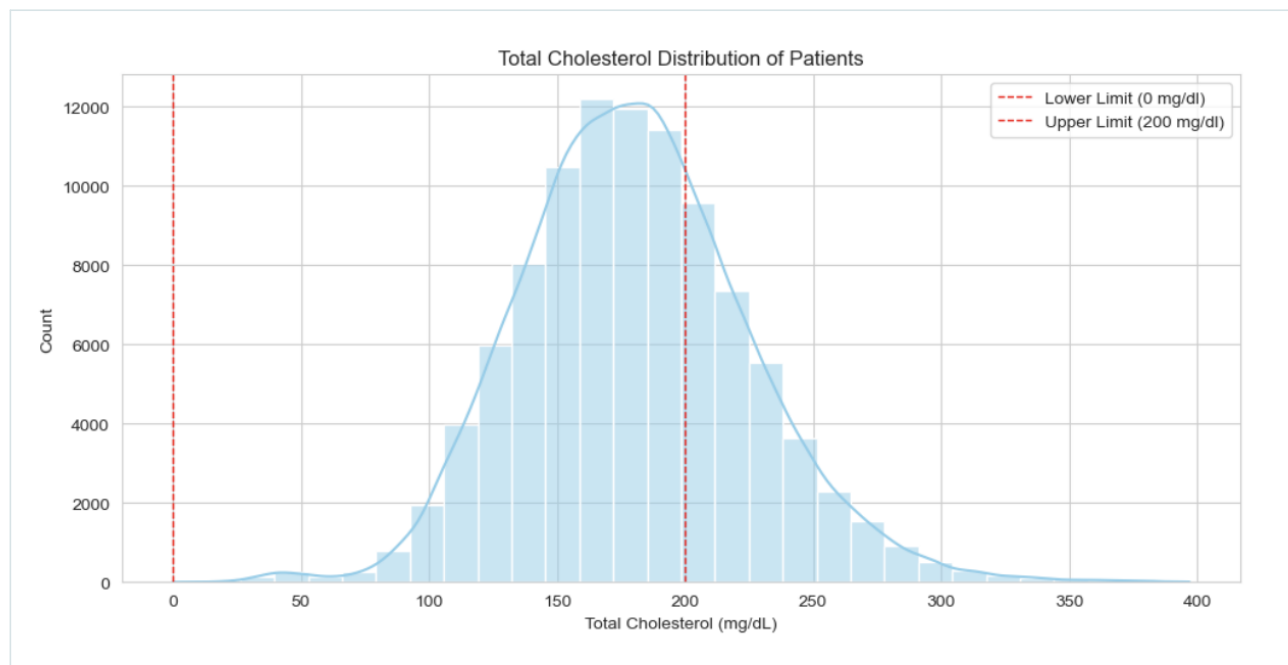


Figure 4 | Total cholesterol, all patients. Dashed line at 200 mg/dL.

The mode sits below the threshold but a substantial shoulder extends past it, and the 200 mg/dL line cuts through populated territory rather than an empty gap.

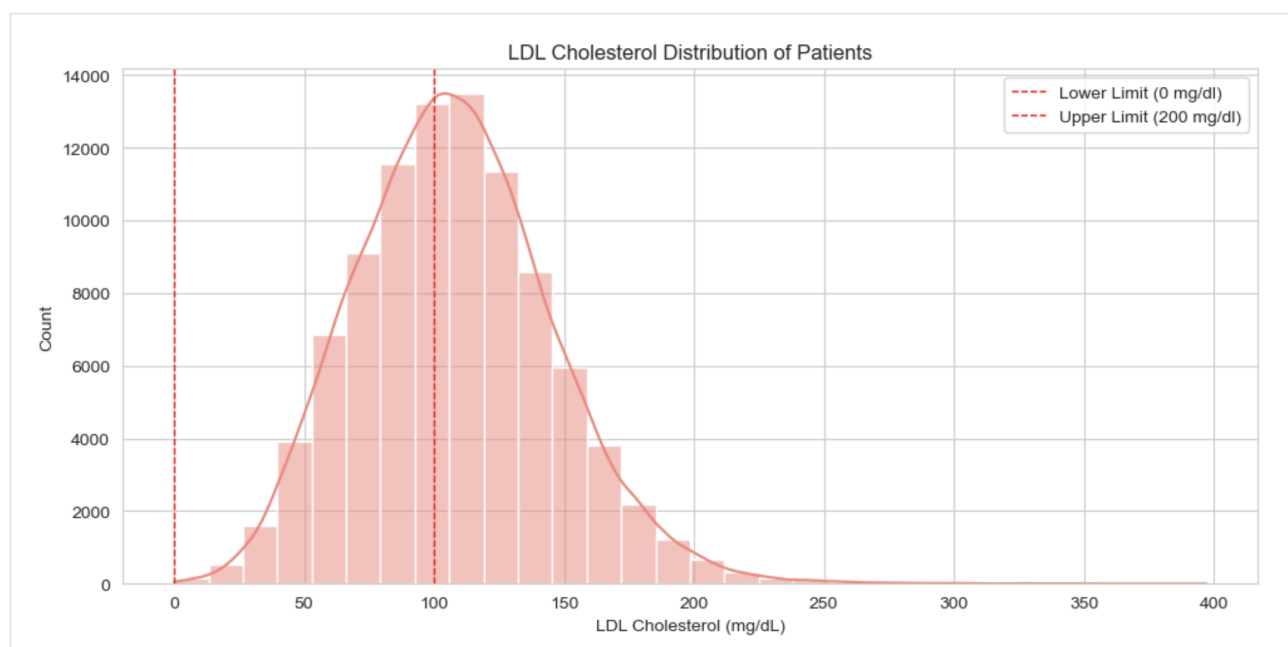


Figure 5 | LDL cholesterol, all patients. Dashed line at 100 mg/dL.

The 100 mg/dL optimal line falls near the centre of the distribution, placing a large share of patients in the near-optimal and above-optimal bands rather than clearly inside or outside the reference range.

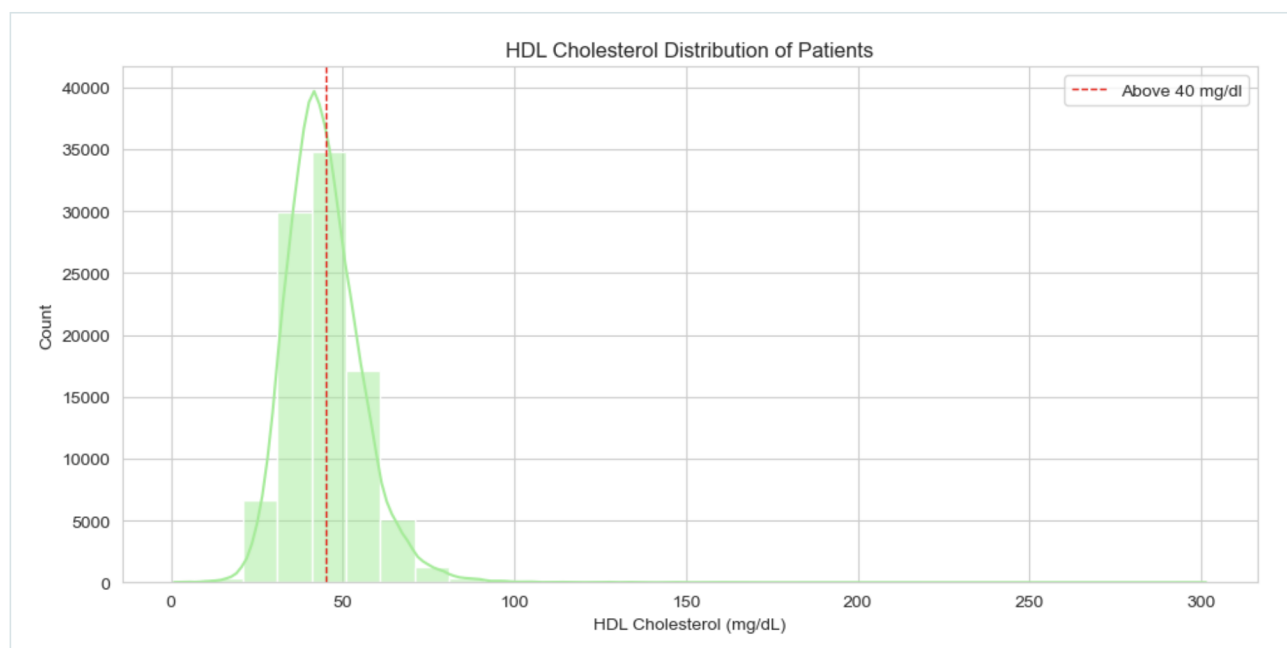


Figure 6 | HDL cholesterol, all patients. Dashed line at 45 mg/dL.

For HDL the desirable direction is reversed: mass to the left of the line is unfavourable. A visible fraction of the cohort sits below 40 mg/dL, the point at which HDL becomes an independent risk factor rather than merely unhelpful.

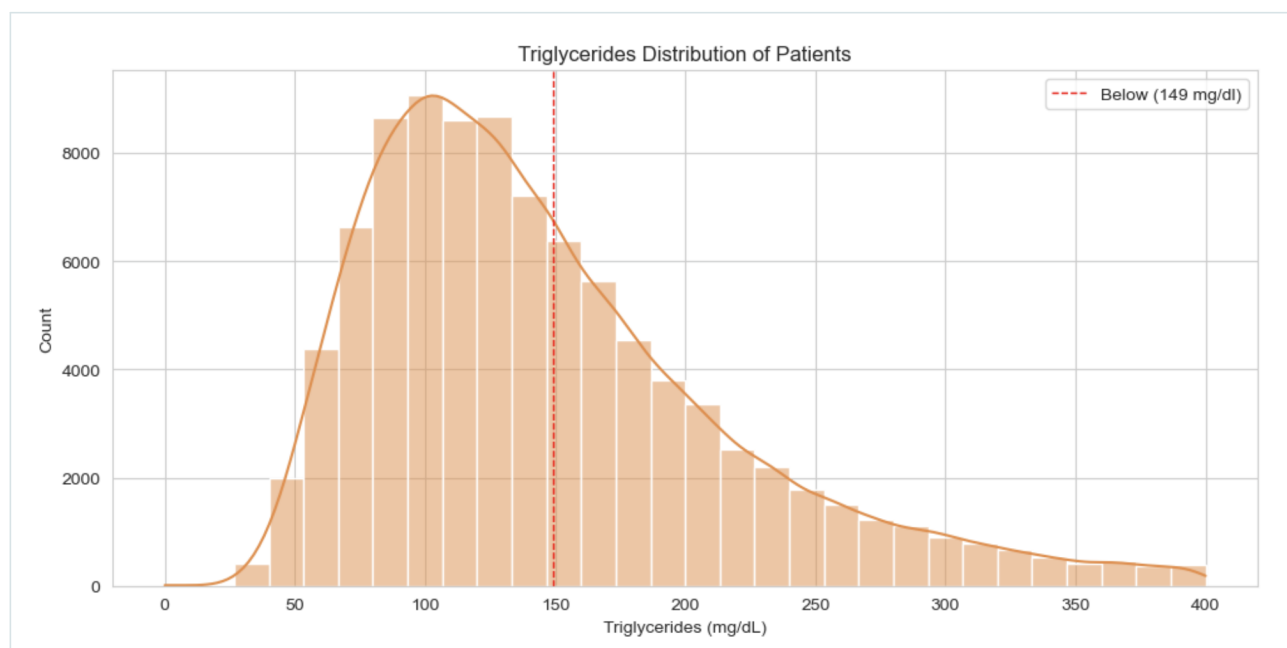


Figure 7 | Triglycerides, all patients. Dashed line at 149 mg/dL.

Triglycerides are the most right-skewed of the four. Because the triglyceride-to-HDL ratio doubles as a marker of insulin resistance, this skew is the point at which the lipid and glucose halves of this report meet.

2.2 Patients under 19 years

Paediatric thresholds are stricter than adult ones for total cholesterol, LDL and triglycerides, and require a higher HDL floor. Only 2,020 patients in the whole cohort fall in the 13–20 band and the under-19 subset is smaller still, so these four distributions are the least stable in the report and should be read as indicative.

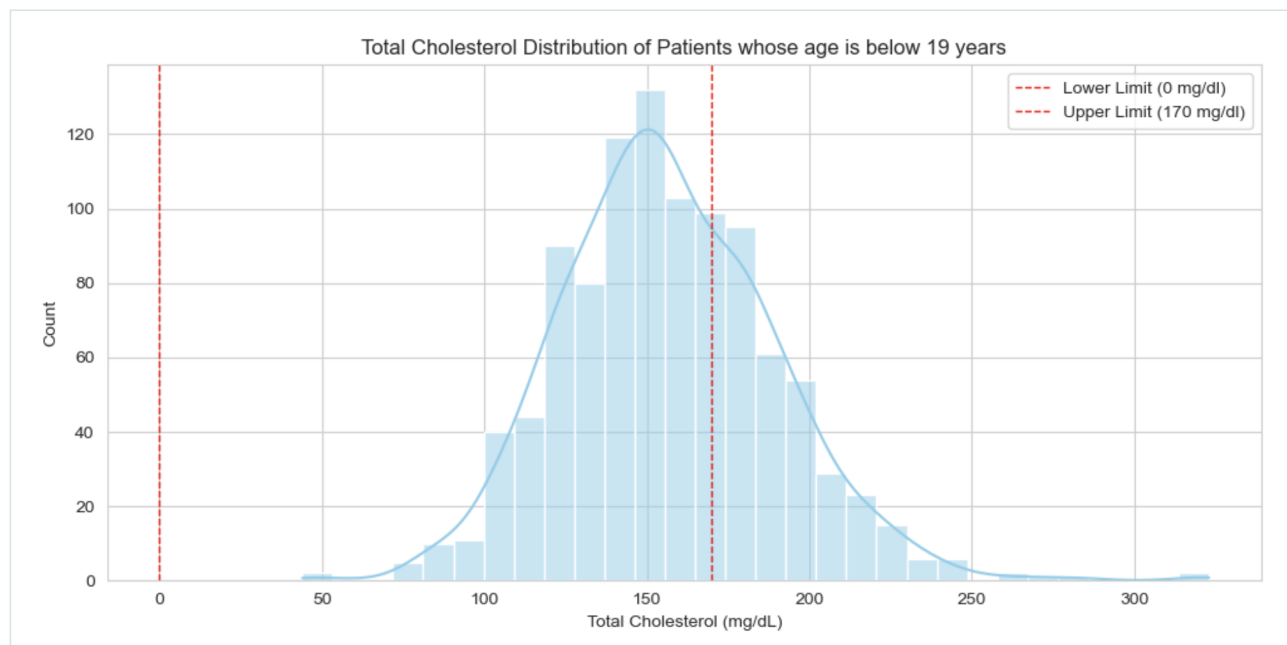


Figure 8 | Total cholesterol, patients under 19. Dashed line at 170 mg/dL.

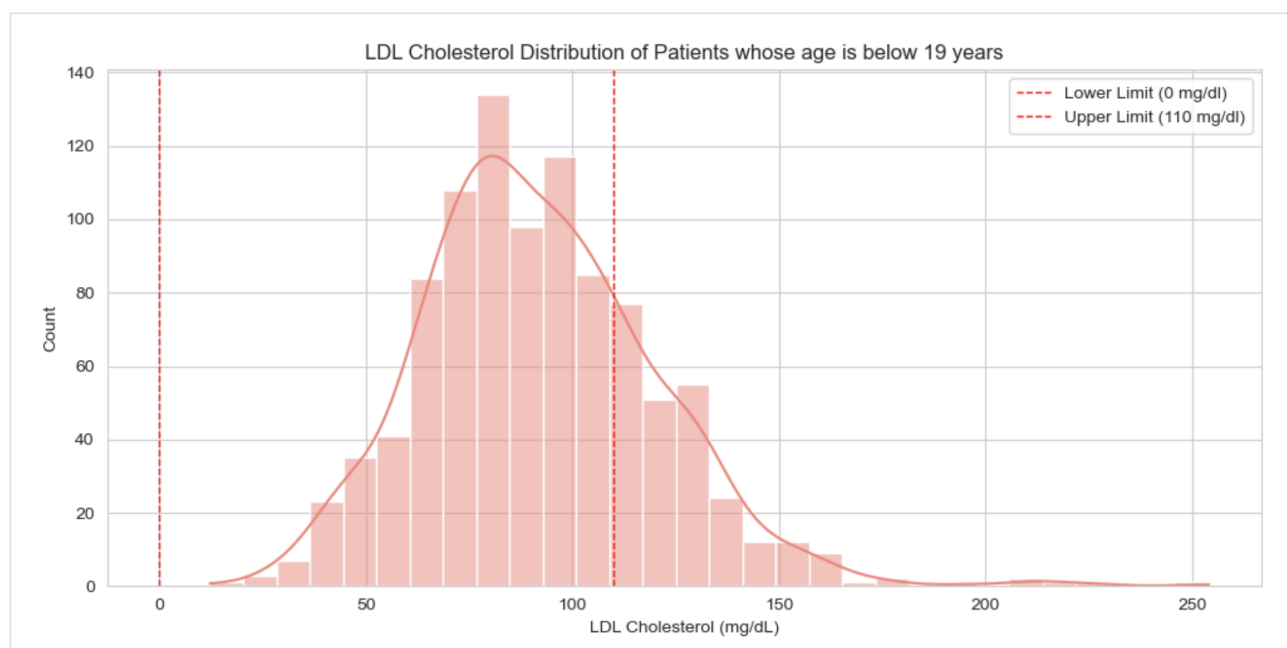


Figure 9 | LDL cholesterol, patients under 19. Dashed line at 110 mg/dL.

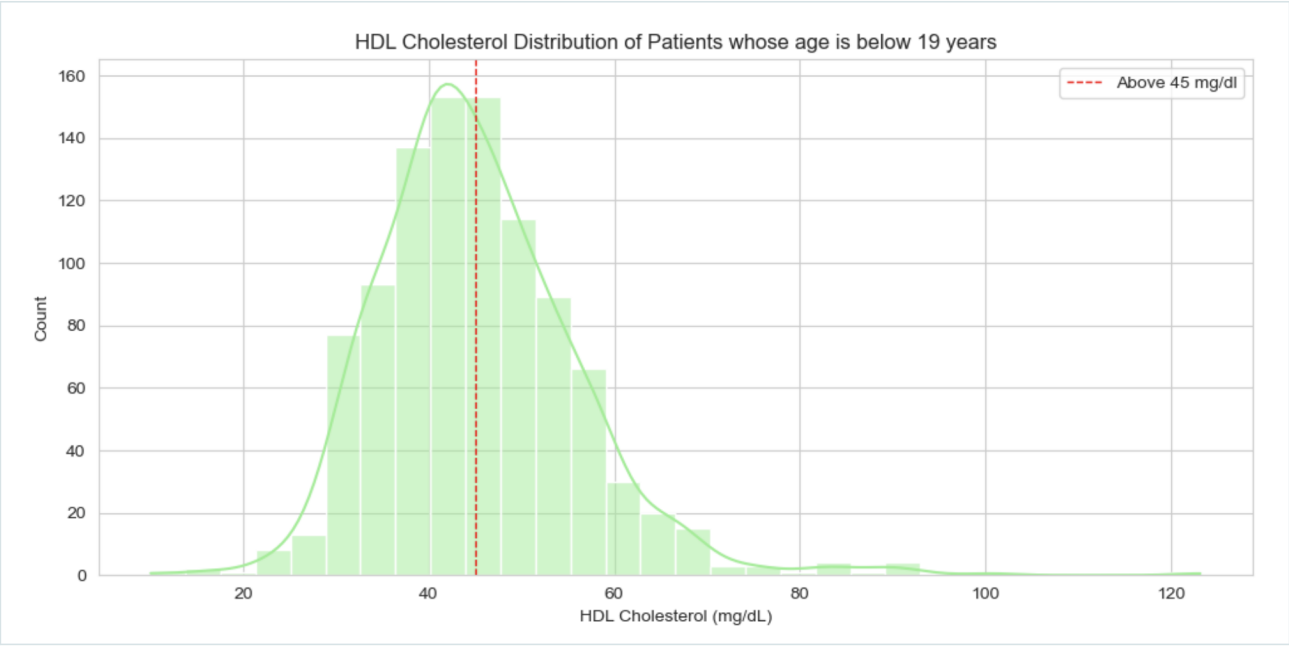


Figure 10 | HDL cholesterol, patients under 19. Dashed line at 45 mg/dL.

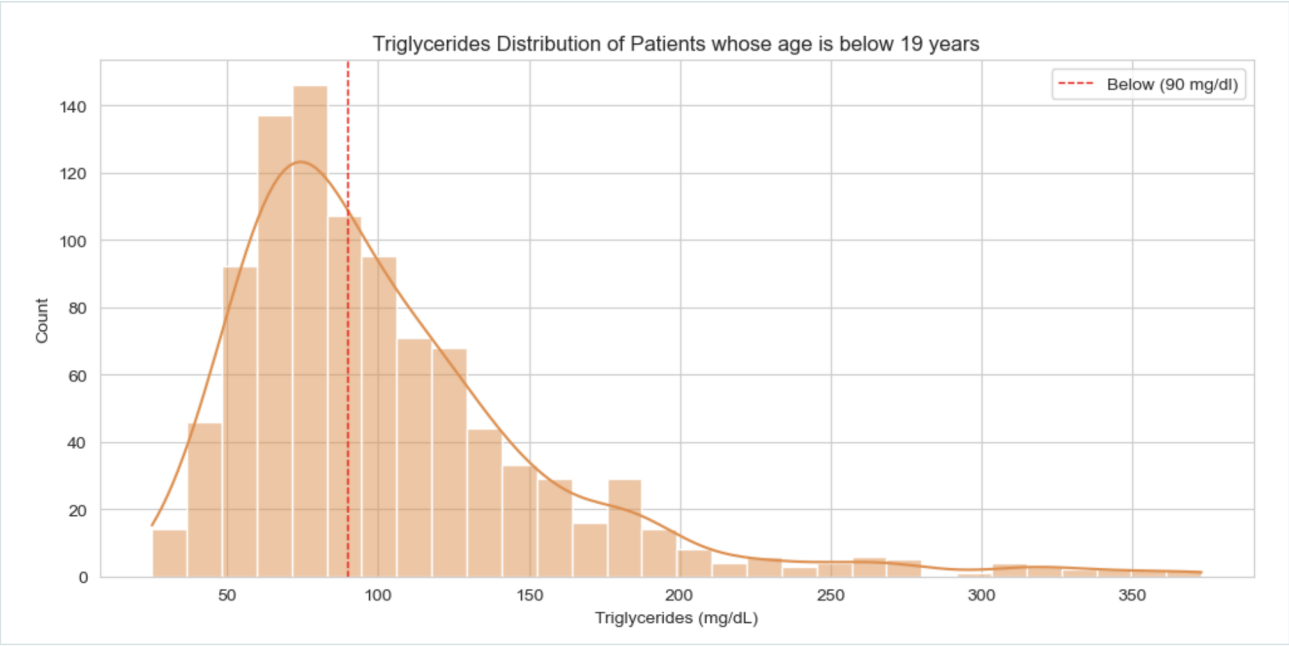


Figure 11 | Triglycerides, patients under 19. Dashed line at 90 mg/dL.

The paediatric triglyceride threshold of 90 mg/dL is the strictest cut in the report, and a clear share of this small group exceeds it — consistent with the 138 prediabetic and 52 diabetic patients found in the same age band in Section 4.

2.3 Male patients, 19 years and over

Adult male thresholds, with the HDL floor at 40 mg/dL. This is the largest of the four cohorts and the one that drives the aggregate distributions in Section 2.1.

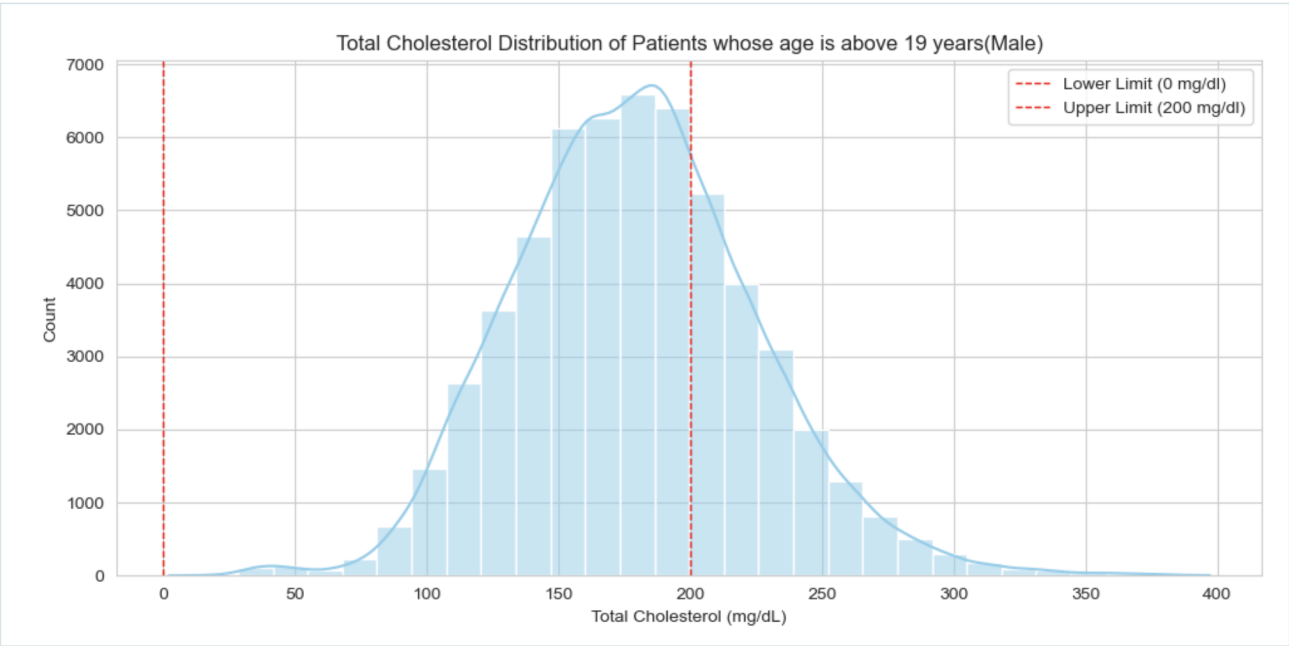


Figure 12 | Total cholesterol, male patients 19 and over. Dashed line at 200 mg/dL.

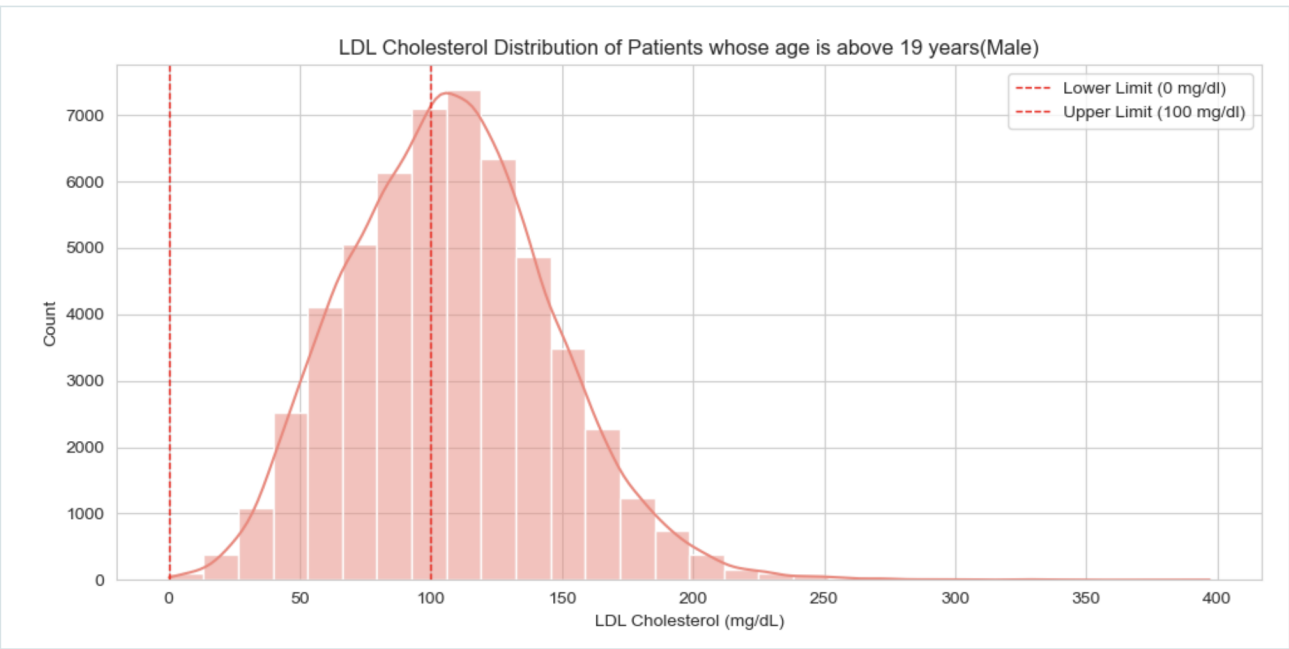


Figure 13 | LDL cholesterol, male patients 19 and over. Dashed line at 100 mg/dL.

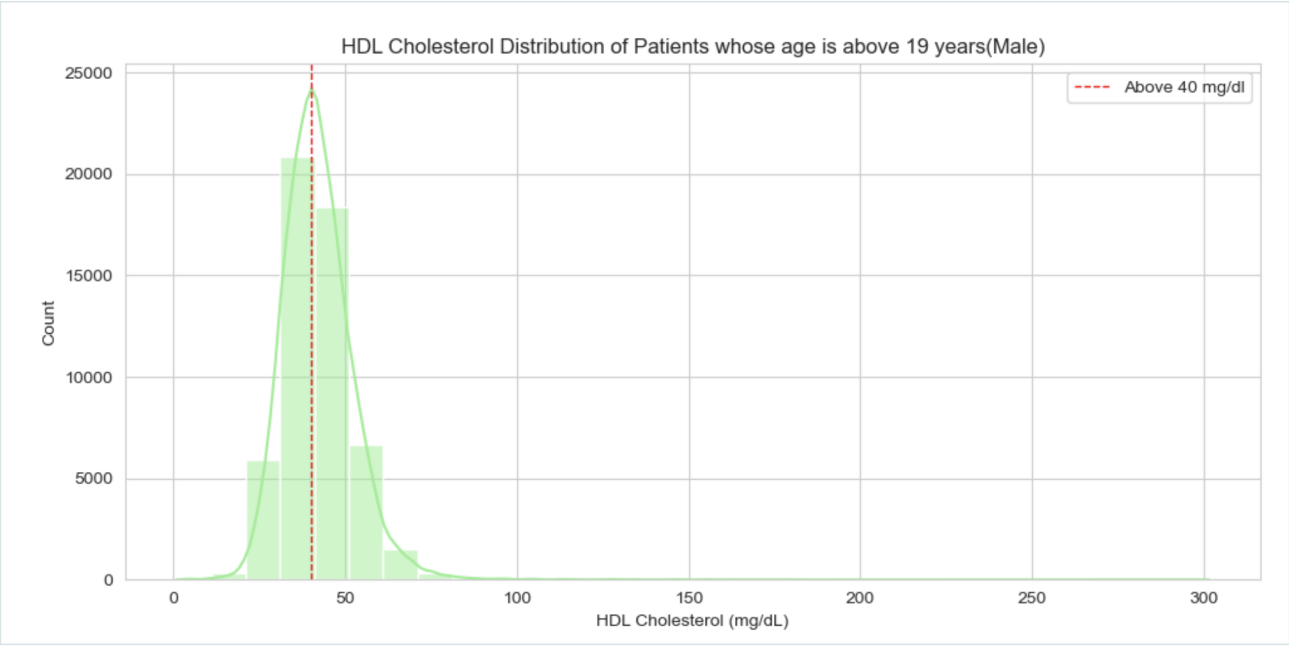


Figure 14 | HDL cholesterol, male patients 19 and over. Dashed line at 40 mg/dL.

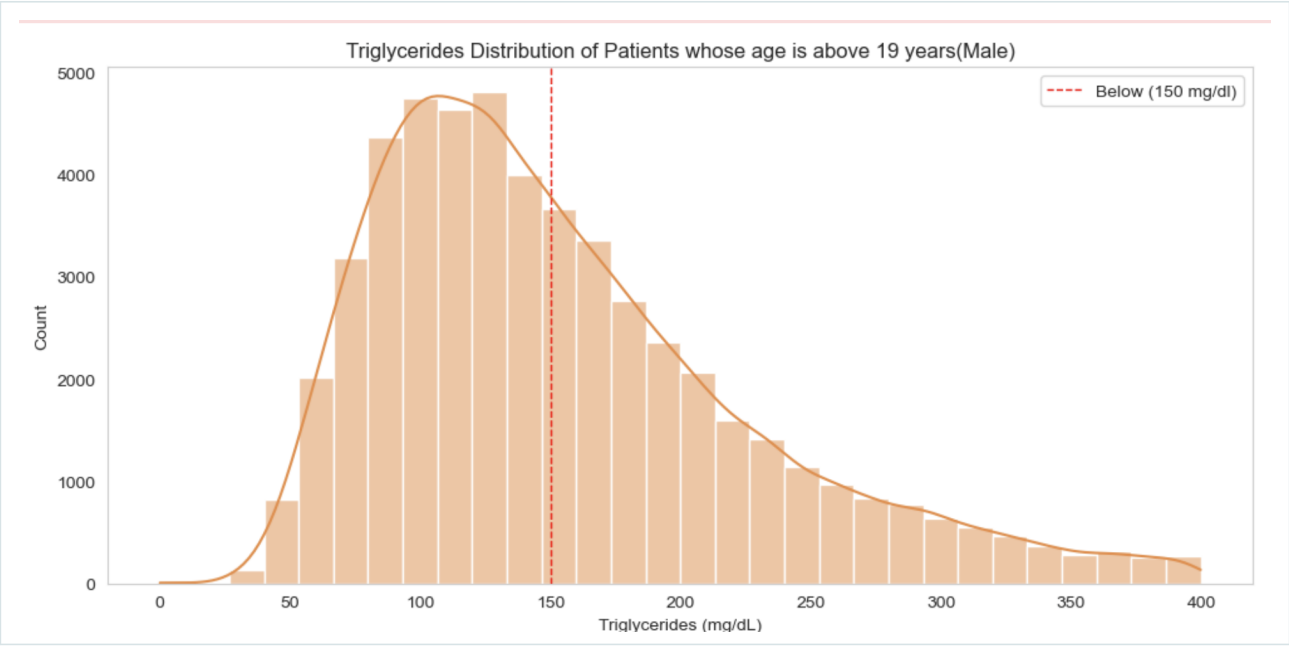


Figure 15 | Triglycerides, male patients 19 and over. Dashed line at 150 mg/dL.

Male triglycerides carry the heavier right tail of the two adult sexes. Taken with the lower HDL floor applied to men, this is the mechanical basis of the 1.54-fold male excess in high-risk ratios reported in Section 3.

2.4 Female patients, 19 years and over

Adult female thresholds. Total cholesterol, LDL and triglycerides use the same cuts as men; the HDL floor is 10 mg/dL higher, at 50 mg/dL.

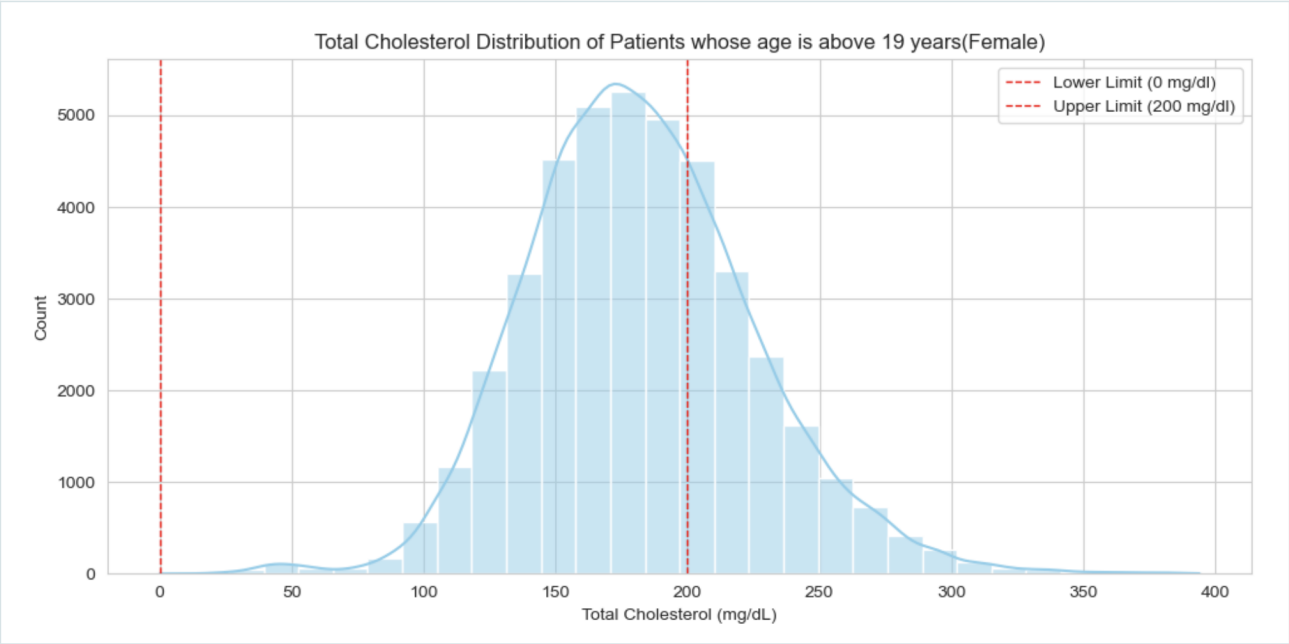


Figure 16 | Total cholesterol, female patients 19 and over. Dashed line at 200 mg/dL.

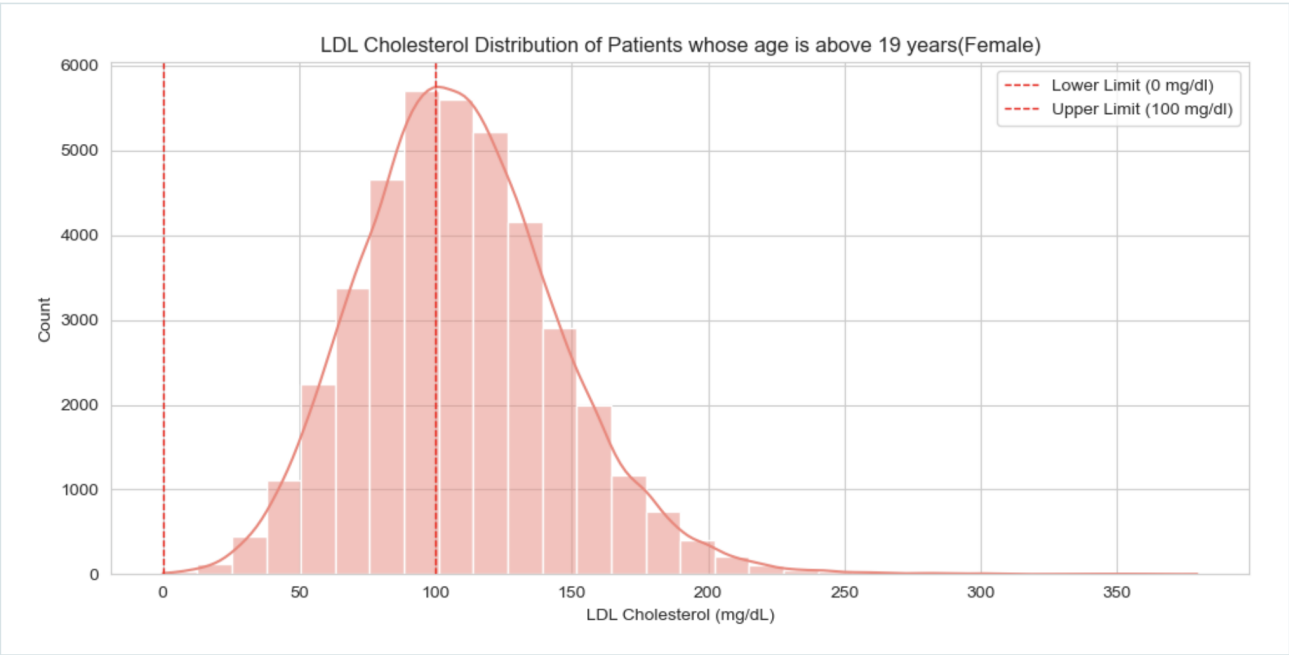


Figure 17 | LDL cholesterol, female patients 19 and over. Dashed line at 100 mg/dL.

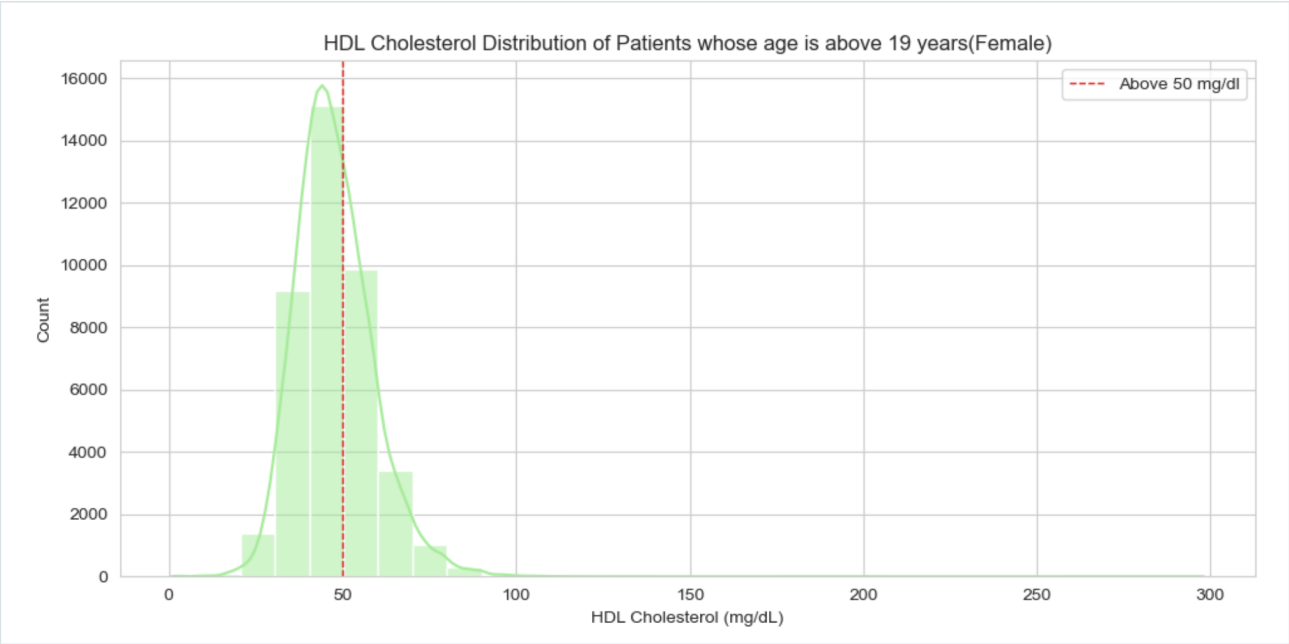


Figure 18 | HDL cholesterol, female patients 19 and over. Dashed line at 50 mg/dL.
Women's HDL distribution sits higher than men's, but it is judged against a higher floor. The stricter cut offsets much of the apparent advantage and is worth remembering when comparing the two cohorts by eye.

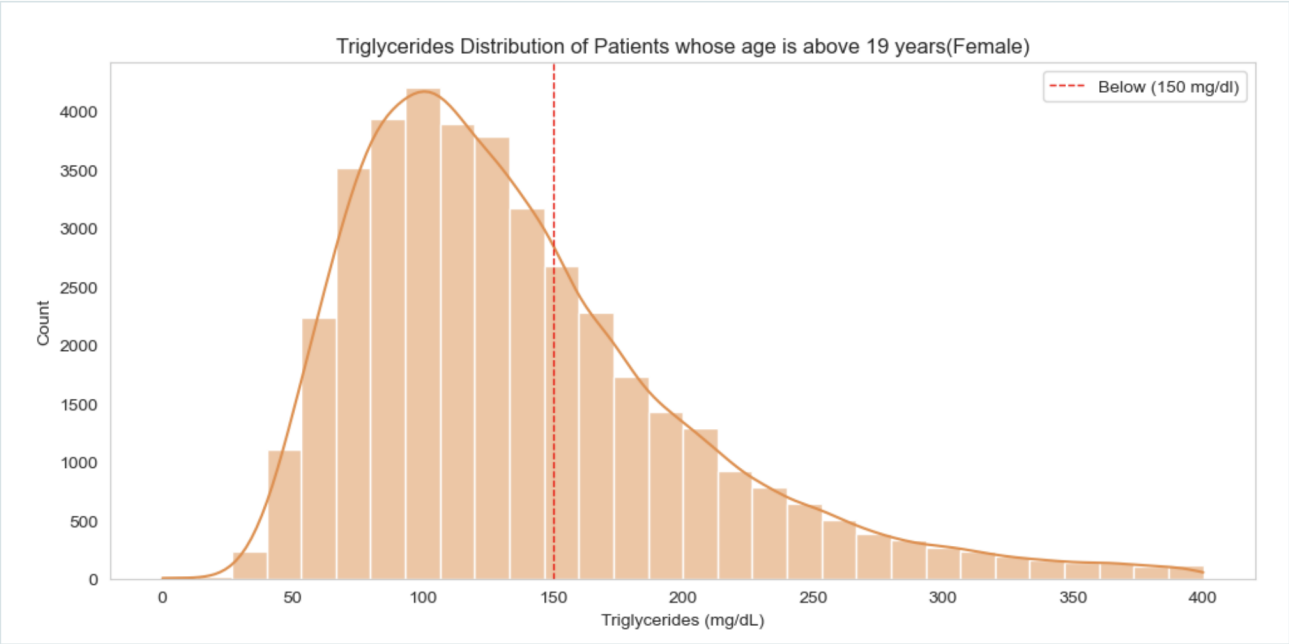


Figure 19 | Triglycerides, female patients 19 and over. Dashed line at 150 mg/dL.

2.5 What the distributions show

Read together, the sixteen histograms make one point that no single chart makes alone. In every cohort and for every analyte, the reference threshold falls inside the body of the distribution rather than in a gap between two populations. There is no bimodality: no separation between a healthy group and an abnormal group. The cohort is a single continuum shifted so that the threshold lands near its middle.

That has a practical consequence. Where a threshold sits in a sparse region, small measurement error moves few patients across it. Where it sits at the mode, as here, a laboratory difference of a few mg/dL reclassifies a large number of people. The borderline population is therefore both large and unstable, and the borderline population is best understood as a structural feature of this cohort rather than as a finding about the individuals inside it.

WHY THE RATIOS MATTER MORE THAN THE LEVELS

A patient with LDL of 130 mg/dL and HDL of 65 mg/dL and a patient with LDL of 130 mg/dL and HDL of 32 mg/dL appear identically abnormal on the LDL histogram and carry very different risk. Single-analyte thresholds cannot see that; ratios can. This is the argument for Section 3, and it is why the ratio-based classification — not the histograms — produces the headline 29.2% figure.

A second consequence concerns the sub-cohorts. Because the HDL floor differs by sex and all thresholds differ for the under-19s, the four sets of histograms are not directly comparable by eye: the same visual position relative to the dashed line means different things in different panels. Section 3 removes that problem by converting everything to a single risk band.

3 Cardiovascular risk ratios

Cholesterol ratios compare the fractions of the lipid panel against one another rather than against a fixed level. Because they are dimensionless, they are robust to the unit chaos documented in Section “Method and data quality”, and because they capture balance rather than magnitude they separate risk more sharply than any single analyte.

Ratio definitions and thresholds

Ratio	Ideal	Acceptable	High risk	What it captures
Total cholesterol / HDL	≤ 3.5	3.5 – 5.0	> 5.0	Total circulating cholesterol against the fraction able to clear it
LDL / HDL	< 2.5	2.5 – 3.5	> 3.5	Plaque-forming against plaque-clearing lipoproteins
HDL / LDL	> 1.0	0.5 – 1.0	< 0.5	The inverse view; higher values are protective
Triglycerides / HDL	< 2.0	2.0 – 3.0	> 3.0	Insulin resistance; the bridge between cardiac and diabetic risk

Table 6 | Ratio thresholds. Each of a patient's ratios was classified independently against its own row, then combined into one band per patient by the majority rule described on the method page.

The triglyceride-to-HDL ratio deserves separate mention. It is the only member of the set that is not purely a cholesterol measure, and it tracks insulin resistance closely enough to serve as a proxy for it. A patient flagged high risk on this ratio alone is carrying metabolic risk that the glucose analysis in Section 4 would also detect — the two sections are not measuring independent things.

Risk distribution across the cohort

Of the 87,761 patients with at least one usable ratio, 25,583 fall in the high-risk band, 34,688 in the acceptable band and 27,490 at ideal levels.

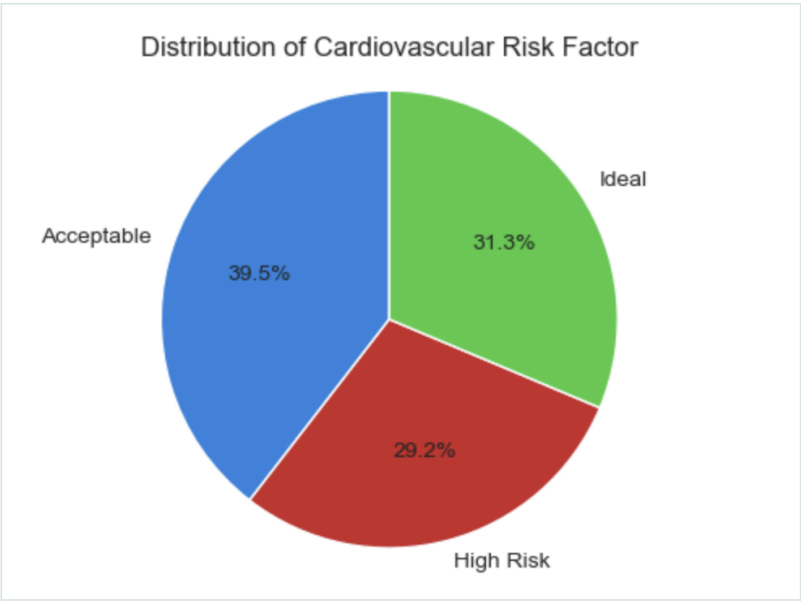


Figure 20 | Distribution of cardiovascular risk bands across the lipid-tested cohort.

Risk band	Patients	Share of 87,761	Clinical reading
Ideal	27,490	31.3%	healthy lipid balance, routine recall
Acceptable	34,688	39.5%	moderate risk, generally reversible by lifestyle
High risk	25,583	29.2%	elevated likelihood of cardiovascular events
Total	87,761	100.0%	

Table 7 | Risk bands. The acceptable band is the largest single group and the one no clinical flag is raised for.

CHECKING THE SHARES

All three bands come from the same 87,761 patients and reconcile to 100.0%: 27,490 ideal (31.3%), 34,688 acceptable (39.5%) and 25,583 high risk (29.2%). A value of 39.2% is sometimes quoted for the acceptable band; it does not reconcile, leaving the three shares short of the whole.

Risk rates by age and sex

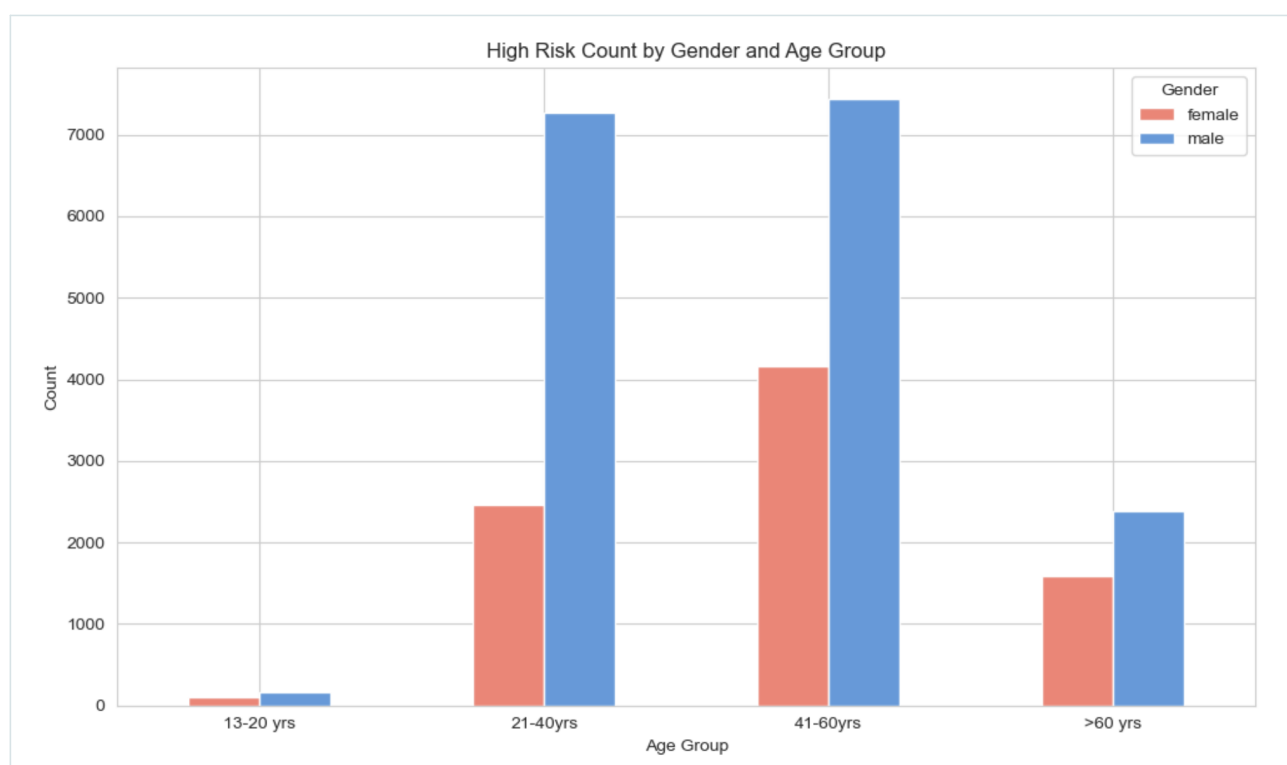


Figure 21 | High-risk patient counts by age band and sex.

Counts peak at 41–60 for both sexes. On its own terms that reading is correct; the rates below qualify it.

Counts, however, follow cohort size. The 41–60 band holds 41,931 patients against 33,922 at 21–40, so it would lead on counts even at a lower rate. Dividing each cell by its denominator from Table 3 changes the conclusion.

Age band	Female n	Female rate	Male n	Male rate	Male : female
13–20 yrs	101	10.7%	167	15.5%	1.45
21–40 yrs	2,456	18.1%	7,265	35.7%	1.97
41–60 yrs	4,167	22.3%	7,444	32.0%	1.43
Above 60 yrs	1,595	16.8%	2,388	18.9%	1.13
All ages	8,319	19.5%	17,264	30.1%	1.54

Table 8 | High-risk lipid rates. Numerators are high-risk patients from Figure 21; denominators are all cohort patients in that cell from Table 3. Because ratios were available for 87.8% of patients, these rates understate the within-tested rate by roughly a factor of 1.14 — about 22.2% for women and 34.3% for men overall. The male-to-female ratio in the final column is unaffected by that scaling.

TWO FINDINGS THE COUNT VIEW HIDES

Male risk peaks at 21–40, not 41–60. 35.7% of men in that band carry a high-risk ratio against 32.0% of men aged 41–60. Women's rate does peak at 41–60, at 22.3%. The single “risk peaks at 41–60” statement is true of counts and of women, and false of male rates.

The sex gap closes with age. The male-to-female rate ratio runs 1.45, 1.97, 1.43, 1.13 across the four bands. The male excess is at its widest in early adulthood and has nearly vanished after 60.

Interpreting the fall after 60

The high-risk rate declines above 60 for both sexes, most steeply for men, from 32.0% to 18.9%. This is unlikely to represent genuinely improving lipid health. Three mechanisms plausibly contribute, and the dataset cannot distinguish between them:

- **Treatment.** Older patients are far more likely to be on lipid-lowering therapy. A statin-controlled ratio is recorded as ideal or acceptable, so effective treatment appears in these data as absence of risk. The dataset holds no medication history, so this cannot be adjusted for.
- **Survivorship.** Patients whose high-risk lipids caused an event before 60 are under-represented among the over-60s who present for routine testing.
- **Testing indication.** Over-60 lipid panels are more often monitoring of a known condition than first-time screening, which selects a differently composed group.

PRACTICAL CONSEQUENCE

The over-60 figure should not be used to de-prioritise older patients. It is the one rate in Table 8 that most likely measures the health system's response to risk rather than the risk itself.

4 Fasting glucose and dysglycaemia

Fasting plasma glucose measures blood sugar after at least eight hours without food. It screens for diabetes and prediabetes, monitors control in known diabetics, and detects insulin resistance early. Results are available for 65,955 patients, 66.0% of the cohort.

Category breakdown

Category	Fasting glucose	Patients	Share of 65,955
Hypoglycaemia	< 70 mg/dL	497	0.8%
Normal	70 – 99 mg/dL	34,215	51.9%
Prediabetes	100 – 125 mg/dL	16,228	24.6%
Diabetes	≥ 126 mg/dL	15,015	22.8%
Total		65,955	100.0%

Table 9 | Fasting glucose categories. Prediabetes and diabetes together – 31,243 patients, 47.4% – are referred to below as dysglycaemia.

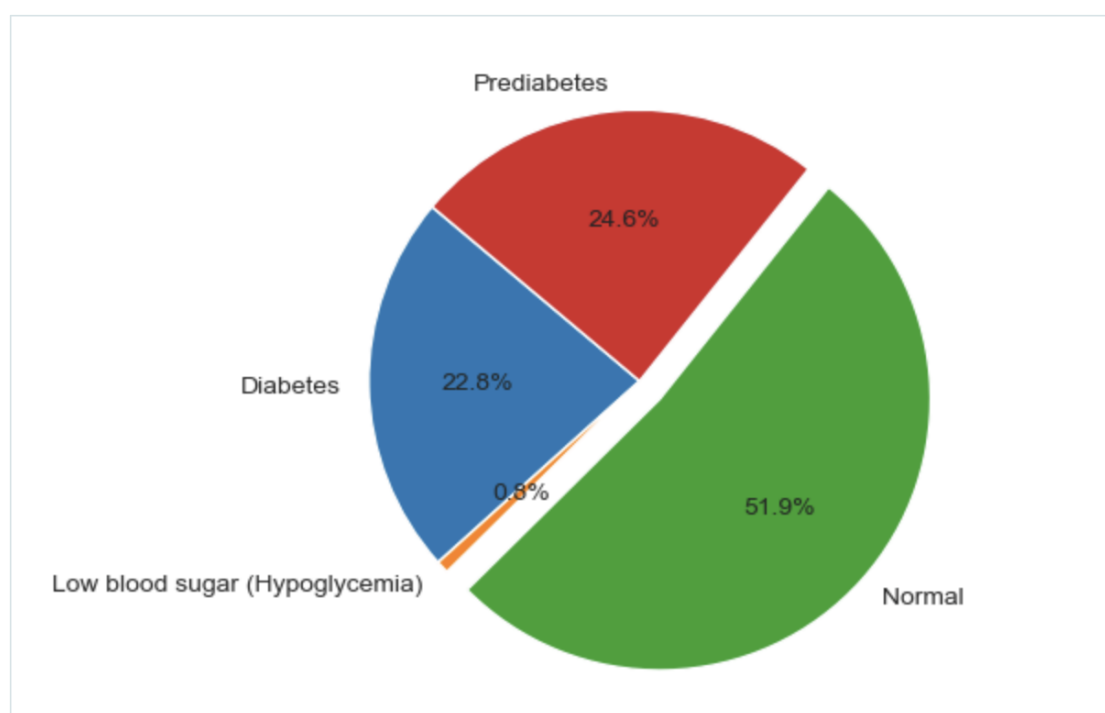


Figure 22 | Share of patients in each fasting-glucose category.

Just over half the tested cohort is normoglycaemic. The prediabetic and diabetic shares are nearly equal in size, at 24.6% and 22.8%.

The hypoglycaemic group

497 patients record a fasting glucose below 70 mg/dL. The group is small but should not be dismissed as noise: nearly 40% of it (198 patients) sits in the 21–40 band, where spontaneous hypoglycaemia is uncommon. In a testing population, low fasting glucose more often reflects over-treatment in a known diabetic or a breach of the eight-hour fasting protocol than a distinct disease. Either explanation is actionable, and neither is visible in the aggregate percentages.

Age and sex gradients

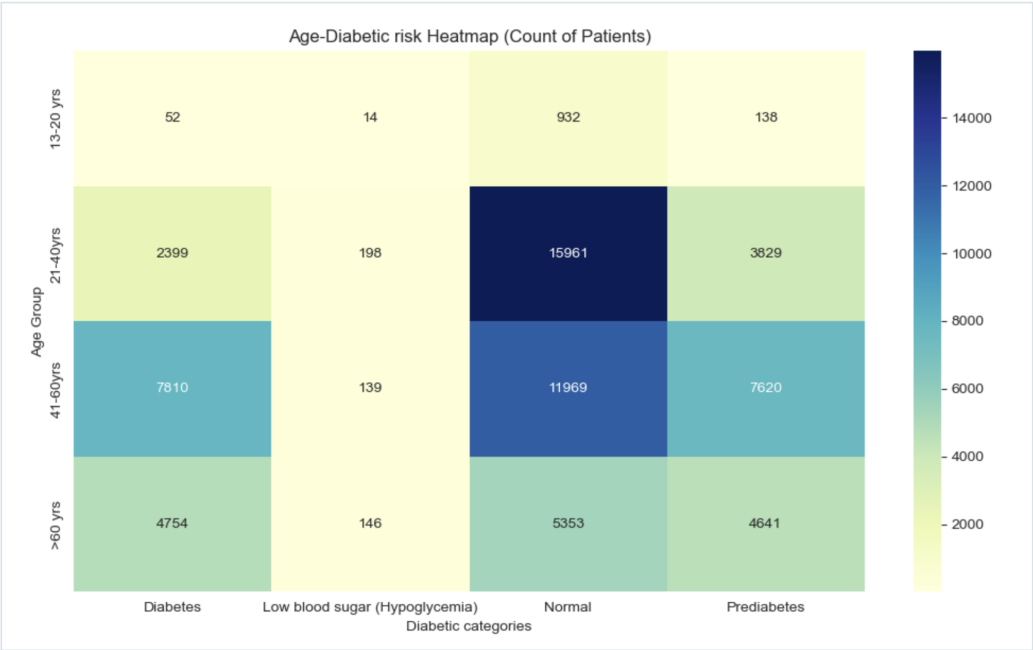


Figure 23 | Patient counts by age band and fasting-glucose category.

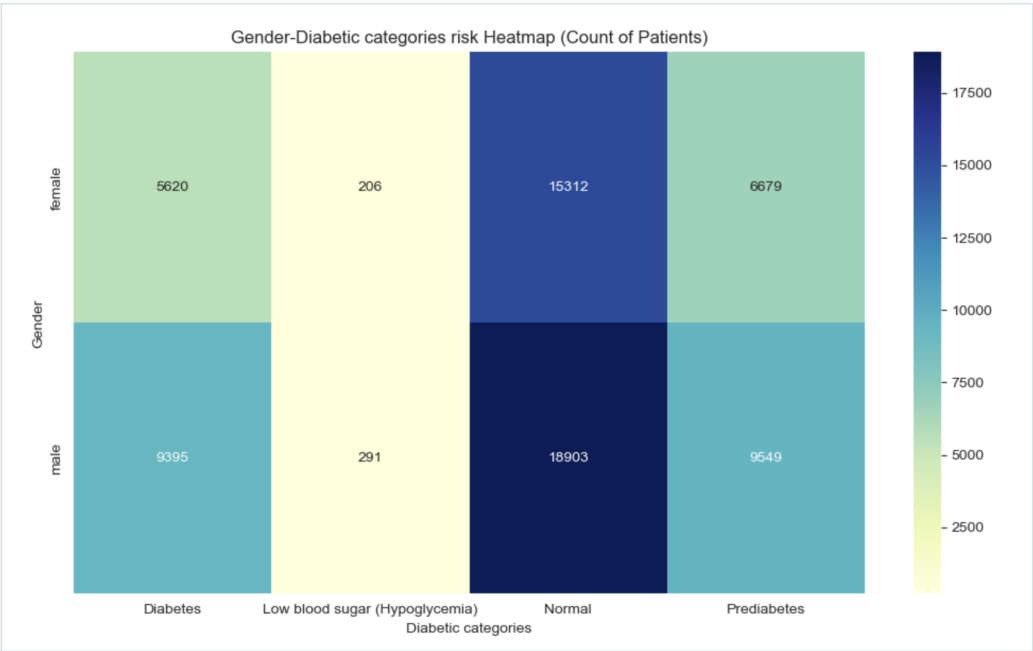


Figure 24 | Patient counts by sex and fasting-glucose category.

Because each row of those heatmaps has its own total, converting to rates is straightforward and needs no assumption about testing coverage. The age gradient is the steepest relationship in this report.

Age band	Tested	Normal	Prediabetes	Diabetes	Dysglycaemic	Prediab : diab
13–20 yrs	1,136	82.0%	12.1%	4.6%	16.7%	2.65
21–40 yrs	22,387	71.3%	17.1%	10.7%	27.8%	1.60
41–60 yrs	27,538	43.5%	27.7%	28.4%	56.0%	0.98
Above 60 yrs	14,894	35.9%	31.2%	31.9%	63.1%	0.98
All ages	65,955	51.9%	24.6%	22.8%	47.4%	1.08

Table 10 | Fasting-glucose rates by age band. Hypoglycaemic patients are included in the tested total but shown in neither the normal nor the dysglycaemic column, so rows do not sum to 100%. Dysglycaemic combines prediabetes and diabetes.

THE CROSSOVER HAPPENS AT 41–60

Diabetes prevalence rises from 4.6% to 10.7% to 28.4% to 31.9% across the four bands — a near threefold jump between 21–40 and 41–60, the largest single step in the report. Within the same band the dysglycaemic share overtakes the normal share, 56.0% against 43.5%. Above 60 the gap widens to 63.1% against 35.9%. From the fifth decade onward, abnormal fasting glucose is the majority condition among patients who get tested.

The final column tracks the conversion pipeline. At 21–40 there are 1.60 prediabetic patients for every diabetic one; by 41–60 the ratio is 0.98 and it stays there. The reservoir of reversible cases is concentrated in young adults, and it drains into diagnosed disease across the fourth and fifth decades.

The sex comparison

Fasting-glucose testing coverage is nearly identical by sex — 66.5% of men and 65.2% of women — so this is the cleanest sex comparison available in the dataset.

Sex	Tested	Coverage	Normal	Prediabetes	Diabetes	Dysglycaemic
Female	27,817	65.2%	55.0%	24.0%	20.2%	44.2%
Male	38,138	66.5%	49.6%	25.0%	24.6%	49.6%
Male : female			0.90	1.04	1.22	1.12

Table 11 | Fasting-glucose rates by sex. Coverage is the tested count as a share of all cohort patients of that sex (42,654 women, 57,346 men).

Men carry the higher burden, but the margin is modest: a diabetes rate 1.22 times the female rate and a dysglycaemia rate 1.12 times it. Prediabetes is almost evenly split at 1.04. Set against the 1.54-fold male excess in high-risk lipid ratios, this is a materially smaller gap, and it is the basis of the fifth finding in the executive summary.

WHAT EQUAL COVERAGE DOES AND DOES NOT FIX

Near-identical testing coverage means the male and female glucose rates are measured on comparably selected groups, so the 1.22 ratio is unlikely to be an artefact of who gets tested. It does not make either rate a population rate. Both sexes are drawn from the same self-selected testing population, so both figures carry the same upward bias — equal coverage makes the comparison sound while leaving the levels overstated.

5 Cross-cutting findings

Side by side, the two halves of this report show three patterns neither reveals alone.

The sex gap is marker-specific

Marker	Female rate	Male rate	Male : female	Denominator
High-risk lipid ratio	19.5%	30.1%	1.54	all cohort patients
Diabetes, FPG ≥ 126	20.2%	24.6%	1.22	glucose-tested patients
Dysglycaemia, FPG ≥ 100	44.2%	49.6%	1.12	glucose-tested patients

Table 12 | Sex gaps compared. The lipid row uses whole-cohort denominators and the glucose rows tested-cohort denominators, so the rate columns are not comparable across rows. The ratio column is, being invariant to a denominator applied equally to both sexes.

Lipid risk is where men and women differ most; glucose dysregulation is close to sex-neutral here. Treating male sex as a general risk marker over-targets men for glucose testing and, more importantly, under-targets women for it.

The two risk axes peak in different decades

Male lipid risk is at its maximum rate by 21–40 and declines thereafter, while dysglycaemia rises monotonically and does not turn over at all within the observed range. The two curves are close to opposite in shape.

Age band	High-risk lipid, male	High-risk lipid, female	Dysglycaemic, both sexes
13–20 yrs	15.5%	10.7%	16.7%
21–40 yrs	35.7%	18.1%	27.8%
41–60 yrs	32.0%	22.3%	56.0%
Above 60 yrs	18.9%	16.8%	63.1%

Table 13 | Age profiles of the two risk axes; bold marks each column's maximum. Denominators as in Table 12.

WHAT FOLLOWS FOR SCREENING

Lipid screening earns its keep earliest, in the twenties and thirties, and for men in particular. Glucose screening earns its keep from 40 onward and keeps earning it. Running both off the same age trigger — the common practice — misses the lipid window and reaches the glucose window on time.

Where the two axes meet

The triglyceride-to-HDL ratio in Section 3 is a recognised proxy for insulin resistance, so a patient flagged high risk on it carries metabolic risk of the kind Section 4 measures directly. The two sections therefore overlap by an amount the release cannot quantify, since identifiers cannot be joined across the two cohorts with confidence about test timing. Measuring that overlap is the most valuable extension available here.

Conclusion and recommendations

Across 100,000 patients, this cohort shows a cardiometabolic burden that is heavy, concentrated in working-age adults, and largely sitting in bands where intervention still changes outcomes. 29.2% of the lipid-tested cohort carries a high-risk cholesterol ratio; 47.4% of the glucose-tested cohort is prediabetic or diabetic. Neither figure describes India, and both describe a population large enough that the patterns inside it are worth acting on.

The distributions in Section 2 explain why the burden is easy to under-read. In every cohort and for every analyte, the reference threshold falls in the middle of the mass rather than in a gap. Most patients are not far from a line; a great many are just on one side of it. The population sits at the threshold, not beyond it, which is the strongest available argument for screening intervals short enough to catch movement rather than only to catch disease.

Recommendations

Three tiers, ordered by how quickly each pays back.

Tier	Population	Size	Action
1 — Immediate	High-risk lipid ratio, or fasting glucose ≥ 126 mg/dL	25,583 + 15,015	Confirmatory repeat testing, then a treatment pathway. A single fasting glucose above 126 mg/dL is a screening signal, not a diagnosis.
2 — Preventive	Acceptable-band lipid ratio, or prediabetic glucose	34,688 + 16,228	Structured lifestyle programme with 6–12 month recall. This is the largest group in the dataset and the one for which no clinical flag currently exists.
3 — Programme design	Screening policy itself	—	Begin lipid screening for men in the late twenties rather than at 40; weight glucose screening toward 40 and above; add a reflex HbA1c for any fasting glucose in the 100–125 mg/dL band.

Table 14 | Recommended actions by tier. Sizes are patient counts from Tables 7 and 9 and are not mutually exclusive across the two markers.

For future data collection

- **HbA1c alongside fasting glucose.** A single fasting measurement cannot distinguish sustained hyperglycaemia from an isolated reading, and cannot confirm diabetes on its own.
- **Medication history.** Without it, the fall in lipid risk above 60 cannot be separated from the effect of treatment, and every age-based lipid finding carries that ambiguity.
- **Geography and collection date.** Neither is present. Both would open regional and temporal analysis, which is where public-health value in a dataset of this size mostly lies.
- **Anthropometry.** Waist circumference or BMI would let the triglyceride-to-HDL signal be interpreted as metabolic syndrome rather than inferred as insulin resistance.
- **Canonical units at the point of entry.** Fifty unit spellings for one analyte is an upstream problem with a cheap upstream fix, and it is the largest single source of avoidable error in this analysis.

Limitations

Every figure in this report is subject to the constraints below. They are stated at length because the temptation to read laboratory data as population data is strong and the error is large.

- **Not a population sample.** The cohort is people who obtained diagnostic testing. That group over-represents the symptomatic, the health-seeking, the insured and the urban. Prevalence figures here will exceed true population prevalence, probably by a wide margin, and the direction of the bias is known while its size is not.
- **Cross-sectional, single measurement.** One fasting glucose reading above 126 mg/dL does not diagnose diabetes; clinical criteria require confirmation on a separate day. The same applies to every lipid threshold. Read all categories as screening classifications.
- **No medication or clinical history.** Patients on statins or antidiabetic therapy are indistinguishable from untreated patients. This most likely explains the fall in lipid risk above 60 and cannot be corrected for.
- **No geography, no dates.** The dataset carries neither, so no regional comparison, no seasonality and no trend analysis is possible, and “across India” cannot be qualified any further than that.
- **Different denominators per marker.** 87.8% of patients have a lipid ratio and 66.0% a fasting glucose. Cross-marker comparisons are indicative only, and the missingness is not random.
- **Conservative risk assignment.** The majority-vote rule with a tie-break toward the more severe band pushes the high-risk share upward. A patient with one high-risk and one ideal ratio is counted as high risk.
- **Residual unit and outlier error.** Harmonising 50 unit spellings and trimming at 400 mg/dL removes gross errors but not subtle ones, and the trim also removes genuine extreme values. Patients with real severe hypertriglyceridaemia are absent from these distributions by construction.
- **Sex recorded as binary.** The source data offers two categories, so this report can offer no more.
- **Small paediatric subset.** Only 2,020 patients fall in the 13–20 band and the under-19 lipid subset is smaller. Those four distributions carry wide uncertainty and no rate is quoted from them.
- **Unquantified overlap between the two risk axes.** The triglyceride-to-HDL ratio and fasting glucose both respond to insulin resistance. The lipid and glucose findings therefore describe overlapping groups by an amount this dataset cannot measure.

STATUS OF THIS REPORT

Exploratory analysis for research and educational purposes. Not a clinical guideline, not a diagnostic instrument, and not a basis for individual patient management.

Appendix A · Reference tables

Raw counts underlying every derived figure in this report, reproduced so that any rate quoted above can be recomputed independently.

A.1 Source and scope

Attribute	Value
Dataset	NidaanKosha-100k-V1.0
Provider	Eka Care
Source	huggingface.co/datasets/ekacare/NidaanKosha-100k-V1.0
Test records	6,844,304
Unique patients	100,000
Age range	17 – 105 years
Fields	document_id, age, gender, test_name, display_ranges, value, unit, specimen, loinc
Tools	Python, pandas, numpy, matplotlib, seaborn, scikit-learn, kmodes

A.2 High-risk lipid counts by age and sex

Age band	Female	Male	Total	Male share
13–20 yrs	101	167	268	62.3%
21–40 yrs	2,456	7,265	9,721	74.7%
41–60 yrs	4,167	7,444	11,611	64.1%
Above 60 yrs	1,595	2,388	3,983	60.0%
All ages	8,319	17,264	25,583	67.5%

Men are 57.3% of the cohort but 67.5% of high-risk patients, an over-representation of 10.2 percentage points.

A.3 Fasting-glucose counts by age band

Age band	Hypoglycaemia	Normal	Prediabetes	Diabetes	Total
13–20 yrs	14	932	138	52	1,136
21–40 yrs	198	15,961	3,829	2,399	22,387
41–60 yrs	139	11,969	7,620	7,810	27,538
Above 60 yrs	146	5,353	4,641	4,754	14,894
All ages	497	34,215	16,228	15,015	65,955

A.4 Fasting-glucose counts by sex

Sex	Hypoglycaemia	Normal	Prediabetes	Diabetes	Total
Female	206	15,312	6,679	5,620	27,817
Male	291	18,903	9,549	9,395	38,138
All	497	34,215	16,228	15,015	65,955

A.5 LOINC codes used

Analyte or ratio	LOINC code(s)
Total cholesterol	2093-3, 14647-2
LDL cholesterol	2089-1, 22748-8
HDL cholesterol	2085-9, 14646-4
Triglycerides	2571-8
Total cholesterol / HDL ratio	9830-1
LDL / HDL ratio	11054-4
HDL / LDL ratio	16616-5
Triglycerides / HDL ratio	44733-4
Fasting plasma glucose	1558-6, 14771-0

Seventeen distinct LOINC codes in the source data match the string “cholesterol”; those above are the codes retained for analysis.